



Air Quality Situation Report

Edmonton Symphony Orchestra Outdoor Concert · Hawrelak Park (Heritage Amphitheatre)

Sat, Sep 5, 2026 · 1:37 PM MDT

Air Quality Health Index (AQHI) — Edmonton

1.0

Current AQHI at Edmonton McCauley (4.75 km from venue)
Forecast Sat 3:00 PM at Edmonton McCauley (4.75 km from venue): **1.1**

Health Risk	AQHI	At-Risk Population	General Population
Low Risk	1.1	Enjoy your usual outdoor activities.	Ideal air quality for outdoor activities.

No active Environment Canada weather alerts for the venue.

Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 28.4°C and feels near 29.1°C. Winds are 18 km/h from SW, gusting near 33 km/h.
- The nearest current AQHI is 1 at Edmonton McCauley, 4.8 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 1 (forecast tonight: 2, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 4.0 km away) puts the area near 1.
- The nearest air monitoring station (Edmonton Lendrum, 3.5 km) reports fine particulate matter (PM2.5) at 3.5 µg/m³.
- A nearby community sensor ('ACA_LaurierHeightsCL', 1.8 km) reads 2.4 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 51.4 km to the NE, i.e. arriving from the NE.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-09-04) is 123.8 km SW of the venue. This fire is roughly upwind of the venue, so smoke transport toward the site is plausible.
- The AQHI forecast for Sat 3:00 PM is 1, based on Edmonton McCauley, 4.8 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Thu 6:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are heat, precipitation, thunderstorm.

Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Increase hydration, shade, cooling and heat-illness messaging.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

Hazard Snapshot — Overall Risk: High Risk

AIR QUALITY Low Risk 1.1 AQHI	HEAT High Risk 33.2 °C apparent/humidex	WIND Moderate Risk 64 km/h peak gust	PRECIPITATION High Risk 5.2 mm/h maximum	THUNDERSTORM High Risk Thu 6:00 PM first forecast signal
AQHI RATE OF CHANGE Low Risk 0.0 AQHI change/hour			WIND SHEAR Low Risk 1.9 ° direction diff (10m vs 120m)	

Local Monitoring

Nearest station 1.0 Edmonton McCauley · 4.75 km	Blended estimate 1.0 confidence High	PM2.5 (station) 3.5 µg/m³ Edmonton Lendrum · 3.51 km	PM2.5 (community) 2.4 µg/m³ ACA_LaurierHeightsCL · 1.8 km
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Prepared by Alberta Capital Airshed for Edmonton Symphony Orchestra outdoor concerts. Beta decision-support product
— for the interactive map, live cameras and hourly outlook, scan the QR code or visit [https://dkevinm.github.io/](https://dkevinm.github.io/eso_sitre/)
eso_sitre/

